

Submission B solution to Problem 9

Overview / Strategy

The proof passes from the ordinary diagonal coinvariant algebra with m bosonic alphabets to the one-bosonic-one-fermionic coinvariant algebra SR_n . A diagonal supersymmetry theorem says that the same universal Schur coefficients $c_\lambda(n)$ occur after replacing $s_\lambda(q_1, \dots, q_m)$ by the super Schur function $s_\lambda(q/u)$. In one bosonic and one fermionic variable, every nonempty non-hook shape vanishes, while

$$s_{(a,1^b)}(q/u) = q^a u^b + q^{a-1} u^{b+1}.$$

This gives a triangular relation between the hook generating functions $C_b(q) = \sum_{a \geq 1} c_{(a,1^b)}(n) q^a$ and the coefficients $D_b(q) = [u^b] \text{Hilb}(SR_n; q, u)$.

The second input is the Fields Frobenius formula, proved by Murai–Rhoades–Wilson. After forgetting the S_n -module structure and using the Robinson–Schensted descent correspondence, this formula becomes a sum over permutations $w \in S_n$. Extracting the coefficient of u^b introduces a Gaussian binomial coefficient. The triangular relation is then solved by induction using the q -Pascal identity, and the Gaussian binomial is finally expanded as a generating function for partitions in a rectangle. This yields the stated counting formula for hook coefficients.

1 Definitions and headline theorem

For $m, n \geq 1$, let

$$P_n^{(m)} = \mathbb{C}[x_{i,j} : 1 \leq i \leq n, 1 \leq j \leq m],$$

with the diagonal action of S_n given by $\sigma \cdot x_{i,j} = x_{\sigma(i),j}$. The coinvariant algebra with m sets of n variables is

$$R_n^{(m)} = P_n^{(m)} / \left\langle (P_n^{(m)})_+^{S_n} \right\rangle,$$

where $(P_n^{(m)})_+^{S_n}$ denotes the invariant polynomials with zero constant term. The algebra is m -multigraded by degrees in the m sets of variables, and

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{i_1, \dots, i_m \geq 0} \dim(R_n^{(m)})_{(i_1, \dots, i_m)} q_1^{i_1} \cdots q_m^{i_m}.$$

As in the problem statement, write

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q_1, \dots, q_m)$$

for the known universal Schur expansion, with coefficients $c_{\lambda}(n)$ independent of m .

For a permutation $w = w_1 \cdots w_n \in S_n$, define

$$\text{Des}(w) = \{i \in \{1, \dots, n-1\} : w_i > w_{i+1}\}, \quad \text{des}(w) = |\text{Des}(w)|, \quad \text{maj}(w) = \sum_{i \in \text{Des}(w)} i.$$

For integers $b, r \geq 0$, the notation $\rho \subseteq b \times r$ means that ρ is a partition with at most b parts, each at most r . Its size is $|\rho|$.

Theorem 1 (Original problem: hook coefficients in the universal Hilbert series). *Fix $n \geq 1$. Then*

$$c_{\emptyset}(n) = 1.$$

For every nonempty hook $\lambda = (a, 1^b)$, where $a \geq 1$ and $b \geq 0$,

$$c_{(a, 1^b)}(n) = \# \left\{ \begin{array}{l} w \in S_n, \quad d = \text{des}(w) \geq b+1, \\ (w, \rho) : \quad \rho \subseteq b \times (d-b-1), \\ a = \text{maj}(w) - bd + \binom{b}{2} + |\rho| \end{array} \right\}.$$

When $b = 0$, the rectangle $0 \times (d-1)$ contains only the empty partition, and the formula becomes

$$c_{(a)}(n) = \#\{w \in S_n : \text{des}(w) \geq 1, \text{maj}(w) = a\} \quad (a \geq 1),$$

with the identity permutation accounted for separately by $c_{\emptyset}(n) = 1$. Also,

$$c_{(a, 1^b)}(n) = 0 \quad \text{whenever } b \geq n-1.$$

The proof of Theorem 1 occupies the rest of the document.

2 Cited inputs

We use two external theorems.

First, define the one-bosonic-one-fermionic coinvariant algebra

$$SR_n = \frac{\mathbb{C}[x_1, \dots, x_n] \otimes \bigwedge(\theta_1, \dots, \theta_n)}{\left\langle (\mathbb{C}[x_1, \dots, x_n] \otimes \bigwedge(\theta_1, \dots, \theta_n))_+^{S_n} \right\rangle},$$

where S_n acts diagonally by permuting the x_i 's and the θ_i 's. The variable q records x -degree and u records θ -degree. The diagonal supersymmetry theorem for bosonic-fermionic coinvariant rings [3] implies that the same universal coefficients $c_{\lambda}(n)$ satisfy

$$\text{Hilb}(SR_n; q, u) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q/u), \tag{1}$$

where $s_\lambda(q/u)$ is the one-bosonic-one-fermionic super Schur function.

Second, the Fields Frobenius formula, proved by Murai–Rhoades–Wilson [4], gives the bigraded Frobenius characteristic of SR_n . For a standard Young tableau T , let

$$\text{Des}(T) = \{i : i+1 \text{ appears in a strictly lower row of } T \text{ than } i\},$$

and define $\text{des}(T) = |\text{Des}(T)|$, $\text{maj}(T) = \sum_{i \in \text{Des}(T)} i$. Then

$$\text{grFrob}(SR_n; q, u) = \sum_{T \in \text{SYT}(n)} q^{\text{maj}(T) - \binom{\text{des}(T)+1}{2}} \prod_{i=1}^{\text{des}(T)} (q^i + u) s_{\text{shape}(T)}. \quad (2)$$

Here $\text{SYT}(n)$ denotes the set of all standard Young tableaux with n boxes, and $s_{\text{shape}(T)}$ is the Schur function appearing in the Frobenius characteristic. This is the Schur-tableau form of the $t = 0$ Delta expression; see also [2]. We also use the standard Robinson–Schensted correspondence and its descent property $\text{Des}(Q(w)) = \text{Des}(w)$ for the recording tableau $Q(w)$; see Fulton [1].

3 Forgetting the S_n -module structure

Lemma 2. *The Hilbert series of SR_n is*

$$\text{Hilb}(SR_n; q, u) = \sum_{w \in S_n} q^{\text{maj}(w) - \binom{\text{des}(w)+1}{2}} \prod_{i=1}^{\text{des}(w)} (q^i + u).$$

Proof. Let $f^\mu = |\text{SYT}(\mu)|$. The Frobenius characteristic sends the irreducible S_n -module of shape μ to s_μ . Therefore applying the dimension map

$$s_\mu \longmapsto f^\mu$$

to a graded Frobenius characteristic gives the corresponding Hilbert series. Applying this map to (2) gives

$$\text{Hilb}(SR_n; q, u) = \sum_{\mu \vdash n} \sum_{T \in \text{SYT}(\mu)} f^\mu q^{\text{maj}(T) - \binom{\text{des}(T)+1}{2}} \prod_{i=1}^{\text{des}(T)} (q^i + u).$$

For each tableau $T \in \text{SYT}(\mu)$, the factor f^μ counts a choice of another tableau $P \in \text{SYT}(\mu)$. Thus the right-hand side is a sum over ordered pairs (P, Q) of standard Young tableaux of the same shape, with $Q = T$, weighted by

$$q^{\text{maj}(Q) - \binom{\text{des}(Q)+1}{2}} \prod_{i=1}^{\text{des}(Q)} (q^i + u).$$

By the Robinson–Schensted correspondence, such pairs (P, Q) are in bijection with permutations $w \in S_n$. Under this bijection, $\text{Des}(Q(w)) = \text{Des}(w)$, hence $\text{des}(Q(w)) = \text{des}(w)$ and $\text{maj}(Q(w)) = \text{maj}(w)$. Substituting these equalities into the preceding sum gives

$$\text{Hilb}(SR_n; q, u) = \sum_{w \in S_n} q^{\text{maj}(w) - \binom{\text{des}(w)+1}{2}} \prod_{i=1}^{\text{des}(w)} (q^i + u),$$

as claimed. □

4 The one-bosonic-one-fermionic specialization

Lemma 3. *With the convention*

$$s_\lambda(q/u) = \sum_{\nu \subseteq \lambda} s_\nu(q) s_{\lambda'/\nu'}(u),$$

one has

$$s_\emptyset(q/u) = 1.$$

If $\lambda = (a, 1^b)$ is a hook with $a \geq 1$, $b \geq 0$, then

$$s_{(a, 1^b)}(q/u) = q^a u^b + q^{a-1} u^{b+1}.$$

Every nonempty non-hook shape has $s_\lambda(q/u) = 0$.

Proof. For an ordinary Schur function in one variable, $s_\nu(q)$ is nonzero only when ν has at most one row. Indeed, in the semistandard tableau definition, only the entry 1 is available, and a column of length at least two would violate strict increase down columns. Thus $s_{(r)}(q) = q^r$, with $(0) = \emptyset$.

Similarly, a skew Schur function $s_{\alpha/\beta}(u)$ in one variable is nonzero exactly when the skew shape α/β has at most one cell in each column, i.e. is a horizontal strip. In that case the unique semistandard filling uses only 1's and contributes $u^{|\alpha/\beta|}$.

For $\lambda = \emptyset$, the only summand is $\nu = \emptyset$, giving $s_\emptyset(q/u) = 1$.

Now suppose $\lambda \neq \emptyset$. If λ is not a hook, then $\lambda_2 \geq 2$. In any potentially nonzero summand, ν must be a row, say $\nu = (r)$, so $\nu' = (1^r)$ is contained in the first column of λ' . Removing ν' from λ' does not affect the second column of λ' , which has length $\lambda_2 \geq 2$. Therefore λ'/ν' has at least two cells in one column and is not a horizontal strip. Hence every summand is zero.

It remains to compute the hook case. Let $\lambda = (a, 1^b)$, so $\lambda' = (b+1, 1^{a-1})$. Again, ν must be a row (r) with $0 \leq r \leq a$. Removing $\nu' = (1^r)$ from the first column of λ' , the remaining part of that first column has length $a-r$. The skew shape λ'/ν' is a horizontal strip if and only if $a-r \leq 1$, i.e. $r = a$ or $r = a-1$. For $r = a$, the contribution is $q^a u^b$. For $r = a-1$, the contribution is $q^{a-1} u^{b+1}$. These are the only nonzero summands, proving the formula. □

5 Gaussian binomial identities

For integers r, s , define

$$\begin{bmatrix} r \\ s \end{bmatrix}_q = \begin{cases} \sum_{\rho \subseteq s \times (r-s)} q^{|\rho|}, & 0 \leq s \leq r, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 4. *The following identities hold.*

1. *If $0 \leq b \leq d$, then*

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{d-b+1}{2}} \begin{bmatrix} d \\ b \end{bmatrix}_q.$$

If $b > d$, both sides are 0, using the convention above.

2. *For $1 \leq b \leq d$,*

$$\begin{bmatrix} d \\ b \end{bmatrix}_q = \begin{bmatrix} d-1 \\ b \end{bmatrix}_q + q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q.$$

Proof. For the first identity, if $b > d$, no term in the product can contribute u^b , and $\begin{bmatrix} d \\ b \end{bmatrix}_q = 0$. Assume $0 \leq b \leq d$, and set $r = d - b$. To obtain the coefficient of u^b , choose r indices

$$1 \leq t_1 < \cdots < t_r \leq d$$

from which the q^i -term is selected. The corresponding contribution is $q^{t_1 + \cdots + t_r}$. Define

$$\rho_j = t_{r+1-j} - (r + 1 - j) \quad (1 \leq j \leq r).$$

Then $\rho_1 \geq \cdots \geq \rho_r \geq 0$, and $\rho_1 \leq d - r = b$, so $\rho \subseteq r \times b$. Conversely, every partition in $r \times b$ arises uniquely in this way. Moreover,

$$t_1 + \cdots + t_r = 1 + \cdots + r + |\rho| = \binom{r+1}{2} + |\rho|.$$

Thus

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{r+1}{2}} \sum_{\rho \subseteq r \times b} q^{|\rho|}.$$

Conjugation of partitions is a size-preserving bijection between partitions in $r \times b$ and partitions in $b \times r$. Hence

$$\sum_{\rho \subseteq r \times b} q^{|\rho|} = \begin{bmatrix} d \\ b \end{bmatrix}_q,$$

because $r = d - b$. This proves the first identity.

For the second identity, if $b = d$, it reads $1 = 0 + 1$, which is immediate. Assume $1 \leq b < d$. Partitions in the rectangle $b \times (d - b)$ split into two disjoint classes. If the first part is at most $d - b - 1$, the partition lies in $b \times (d - b - 1)$, and these partitions contribute $\begin{bmatrix} d-1 \\ b \end{bmatrix}_q$. If the first part is exactly $d - b$, removing that first part leaves a partition in $(b - 1) \times (d - b)$, and the removed part contributes a factor q^{d-b} . These partitions therefore contribute $q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q$. The two cases are disjoint and exhaustive, proving the q -Pascal identity. $\square$

6 The coefficients of u^b

For $b \geq 0$, define

$$D_b(q) = [u^b] \text{Hilb}(SR_n; q, u).$$

Lemma 5. *For every $b \geq 0$,*

$$D_b(q) = \sum_{w \in S_n} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \left[\begin{matrix} \text{des}(w) \\ b \end{matrix} \right]_q.$$

Proof. Use Lemma 2. Fix $w \in S_n$, and write

$$d = \text{des}(w), \quad M = \text{maj}(w).$$

The contribution of w to $\text{Hilb}(SR_n; q, u)$ is

$$q^{M - \binom{d+1}{2}} \prod_{i=1}^d (q^i + u).$$

By Lemma 4,

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{d-b+1}{2}} \left[\begin{matrix} d \\ b \end{matrix} \right]_q,$$

with the understanding that the expression is zero when $b > d$. Multiplying by the prefactor gives

$$q^{M - \binom{d+1}{2}} q^{\binom{d-b+1}{2}} \left[\begin{matrix} d \\ b \end{matrix} \right]_q.$$

Since

$$-\binom{d+1}{2} + \binom{d-b+1}{2} = -bd + \binom{b}{2},$$

this contribution equals

$$q^{M - bd + \binom{b}{2}} \left[\begin{matrix} d \\ b \end{matrix} \right]_q.$$

Summing over all $w \in S_n$ proves the formula. □

7 Solving the triangular relation

For $b \geq 0$, define the hook generating polynomial

$$C_b(q) = \sum_{a \geq 1} c_{(a, 1^b)}(n) q^a,$$

and set $C_{-1}(q) = 0$.

Proposition 6. *One has $c_{\emptyset}(n) = 1$. Moreover, for every $b \geq 0$,*

$$C_b(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq b+1}} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \begin{bmatrix} \text{des}(w) - 1 \\ b \end{bmatrix}_q.$$

Proof. First, the ideal defining $R_n^{(m)}$ is homogeneous and generated by positive-degree elements, so the degree-zero component of $R_n^{(m)}$ is $\mathbb{C}$. Hence

$$\text{Hilb}(R_n^{(m)}; 0, \dots, 0) = 1.$$

In the universal Schur expansion, $s_{\emptyset} = 1$, while every nonempty Schur polynomial has zero constant term. Therefore $c_{\emptyset}(n) = 1$.

By (1) and Lemma 3,

$$\text{Hilb}(SR_n; q, u) = 1 + \sum_{b \geq 0} \sum_{a \geq 1} c_{(a, 1^b)}(n) (q^a u^b + q^{a-1} u^{b+1}),$$

because all nonempty non-hook shapes vanish under the specialization q/u . Taking the coefficient of u^b gives

$$D_b(q) = \delta_{b,0} + C_b(q) + q^{-1} C_{b-1}(q), \quad (3)$$

where $\delta_{b,0}$ is the Kronecker delta and $C_{-1}(q) = 0$.

We now prove the displayed formula for $C_b(q)$ by induction on b .

For $b = 0$, equation (3) gives

$$C_0(q) = D_0(q) - 1.$$

By Lemma 5,

$$D_0(q) = \sum_{w \in S_n} q^{\text{maj}(w)}.$$

The unique permutation with no descents is $12 \cdots n$, and it contributes 1. Every other permutation has at least one descent. Therefore

$$C_0(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq 1}} q^{\text{maj}(w)}.$$

This is the required formula for $b = 0$, since $\begin{bmatrix} \text{des}(w) - 1 \\ 0 \end{bmatrix}_q = 1$ whenever $\text{des}(w) \geq 1$.

Now let $b \geq 1$, and assume the formula is known for $b - 1$. From (3),

$$C_b(q) = D_b(q) - q^{-1} C_{b-1}(q).$$

Fix $w \in S_n$, and write $d = \text{des}(w)$, $M = \text{maj}(w)$. By Lemma 5, the contribution of w to $D_b(q)$ is

$$q^{M - bd + \binom{b}{2}} \begin{bmatrix} d \\ b \end{bmatrix}_q.$$

By the induction hypothesis, the contribution of w to $q^{-1}C_{b-1}(q)$, when $d \geq b$, is

$$q^{-1}q^{M-(b-1)d+\binom{b-1}{2}} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q.$$

The exponent satisfies

$$-1 + M - (b-1)d + \binom{b-1}{2} = M - bd + \binom{b}{2} + d - b.$$

Thus, for all w with $d \geq b$, the difference of the two contributions is

$$q^{M-bd+\binom{b}{2}} \left(\begin{bmatrix} d \\ b \end{bmatrix}_q - q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q \right).$$

If $d < b$, both contributions are zero. For $d \geq b$, Lemma 4 gives

$$\begin{bmatrix} d \\ b \end{bmatrix}_q - q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q = \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

Therefore

$$C_b(q) = \sum_{w \in S_n} q^{M-bd+\binom{b}{2}} \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

The Gaussian binomial $\begin{bmatrix} d-1 \\ b \end{bmatrix}_q$ is zero unless $d-1 \geq b$, i.e. unless $d \geq b+1$. Hence

$$C_b(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq b+1}} q^{\text{maj}(w)-b \text{des}(w)+\binom{b}{2}} \begin{bmatrix} \text{des}(w)-1 \\ b \end{bmatrix}_q,$$

which completes the induction. □

8 Proof of the headline theorem

Proof of Theorem 1. The equality $c_{\emptyset}(n) = 1$ is part of Proposition 6.

Fix $b \geq 0$. Proposition 6 gives

$$C_b(q) = \sum_{\substack{w \in S_n \\ d=\text{des}(w) \geq b+1}} q^{\text{maj}(w)-bd+\binom{b}{2}} \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

By the definition of the Gaussian binomial coefficient,

$$\begin{bmatrix} d-1 \\ b \end{bmatrix}_q = \sum_{\rho \subseteq b \times (d-b-1)} q^{|\rho|},$$

because $(d - 1) - b = d - b - 1$. Therefore

$$C_b(q) = \sum_{\substack{w \in S_n \\ d = \text{des}(w) \geq b+1}} \sum_{\rho \subseteq b \times (d-b-1)} q^{\text{maj}(w) - bd + \binom{b}{2} + |\rho|}.$$

On the other hand,

$$C_b(q) = \sum_{a \geq 1} c_{(a, 1^b)}(n) q^a.$$

Thus the coefficient of q^a in $C_b(q)$ is exactly the number of pairs (w, ρ) such that

$$\begin{aligned} w \in S_n, \quad d = \text{des}(w) \geq b + 1, \\ \rho \subseteq b \times (d - b - 1), \end{aligned}$$

and

$$a = \text{maj}(w) - bd + \binom{b}{2} + |\rho|.$$

This is precisely the boxed formula in the theorem.

It remains only to verify the stated edge cases. First, the exponent just displayed is always positive under the condition $d \geq b + 1$. Indeed, if the descent set of w has size d , say

$$\text{Des}(w) = \{i_1 < \cdots < i_d\},$$

then $i_j \geq j$ for every j , so

$$\text{maj}(w) = i_1 + \cdots + i_d \geq 1 + \cdots + d = \binom{d+1}{2}.$$

Consequently,

$$\text{maj}(w) - bd + \binom{b}{2} \geq \binom{d+1}{2} - bd + \binom{b}{2} = \binom{d-b+1}{2}.$$

Since $d \geq b + 1$, the last binomial coefficient is at least 1, and adding $|\rho| \geq 0$ keeps the exponent positive.

When $b = 0$, the rectangle $0 \times (d - 1)$ contains only the empty partition. The condition $d \geq b + 1$ becomes $\text{des}(w) \geq 1$, and the equation for a becomes $a = \text{maj}(w)$. Hence

$$c_{(a)}(n) = \#\{w \in S_n : \text{des}(w) \geq 1, \text{maj}(w) = a\} \quad (a \geq 1).$$

The identity permutation has $\text{des} = 0$ and $\text{maj} = 0$, and its contribution is the separate term $c_{\emptyset}(n) = 1$.

Finally, every permutation in S_n has at most $n - 1$ descents. If $b \geq n - 1$, then the required inequality $d = \text{des}(w) \geq b + 1$ is impossible. Therefore

$$c_{(a, 1^b)}(n) = 0 \quad \text{for all } b \geq n - 1.$$

The theorem is proved. □

References

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